

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse structured JSON audit logs, identify, and mask PII fields (Social Security Numbers, Credit Card numbers) before persisting records to storage.	High	Pass: Masked audit log saved with SHA-256 integrity hash. Fail: Cleartext credit card number stored in database.	GDPR requires PII to be protected at the point of ingestion, before any record is durably stored.
FR-002	Functional	The system shall generate a cryptographically verifiable, immutable export of audit records by chaining each record's SHA-256 hash to the hash of the preceding record.	High	Pass: The computed hash chain of an exported file matches the stored chain, and altering any single record in the export is detected on re-verification. Fail: A tampered record passes verification undetected.	Auditors require mathematical proof that the audit trail has not been altered after the fact.
FR-003	Functional	The system shall allow a Compliance Officer to query and filter ingested audit logs by date range, source application, event	Medium	Pass: A filtered query against a seeded 100-record test set returns only records matching all specified criteria. Fail: Any non-matching record appears in the result set.	Enables targeted compliance reviews and incident investigations without manual log inspection.

		type, and user ID.			
FR-004	Functional	The system shall allow a Security Auditor to independently verify the integrity of an exported audit trail by re-computing and validating its hash chain against the original stored hash values.	High	Pass: A filtered query against a seeded 100-record test set returns only records matching all specified criteria. Fail: Any non-matching record appears in the result set.	Provides an independent, auditor-initiated tamper check separate from internal system validation.
FR-005	Functional	The system shall record every administrative action (e.g., configuration change, user role change, masking-rule update) as its own immutable audit event, included in the hash chain.	Medium	Pass: An unmodified export returns "VERIFIED"; an export with a single byte modified returns "TAMPERED" and identifies the affected record ID. Fail: A tampered export returns "VERIFIED".	Prevents privileged users from making untracked changes that could undermine the system's compliance guarantees.
NFR-001	Nonfunctional (Performance & Security)	Audit log query endpoints must support complex filtering across	High	Pass: Performing a configuration change creates a new audit	Compliance and incident-response workflows require

		10 million historical records in under 1 second.		record containing actor ID, timestamp, and action type, and that record is included in the hash chain. Fail: An administrative action produces no corresponding audit record.	near-real-time access to historical logs during audits or breach investigations.
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